
Light

The speed of light is constant

$$c = 3 \times 10^8 \text{ m/s}$$

Ex. How long does it take light to go: 1m?

to the sun? (AU = 1.5×10^{11} m)

How far away in m is andromeda galaxy if we see it 2.2 Myr in the past?

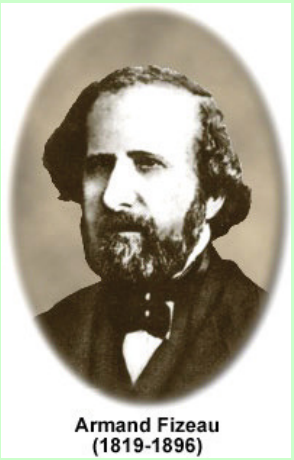
We know because of:

Galileo



Romer

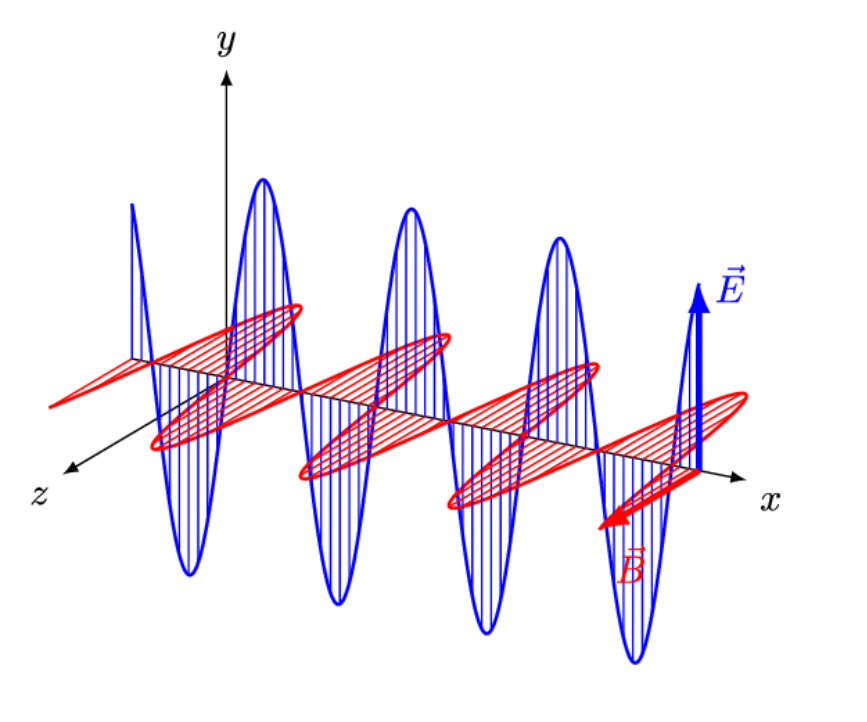




Fizeau

c.f. Sec 1.2 pg 1

What is light? A self propagating EM disturbance



1. Light propagates in the direction of the Poynting Vector, S

$$\vec{S} = \frac{1}{\mu_0}\vec{E} \times \vec{B}$$

2. Light propagates at the speed of light, c

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$

$$\epsilon_0 = 8.85 \times 10^{-12}\ \mathrm{m}^{-3}\ \mathrm{kg}^{-1}\ \mathrm{s}^{4}\ \mathrm{A}^{2}$$

$$\mu_0 = 4\pi \times 10^{-7}\ \mathrm{m}\ \mathrm{kg}\ \mathrm{s}^{-2}\ \mathrm{A}^{-2}$$

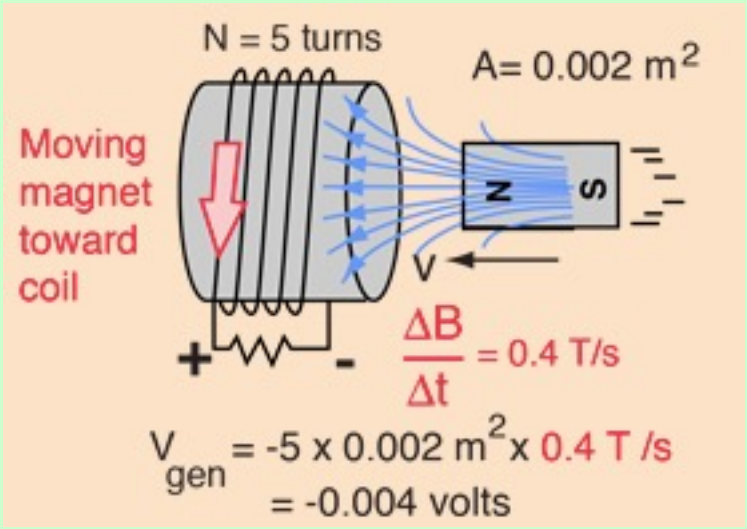
3. The magnetic field, B is always perp. to the electric field, E

$$|\vec{B}| = \frac{|\vec{E}|}{c}$$

Why do EM disturbances propagate?

Demo: oompa

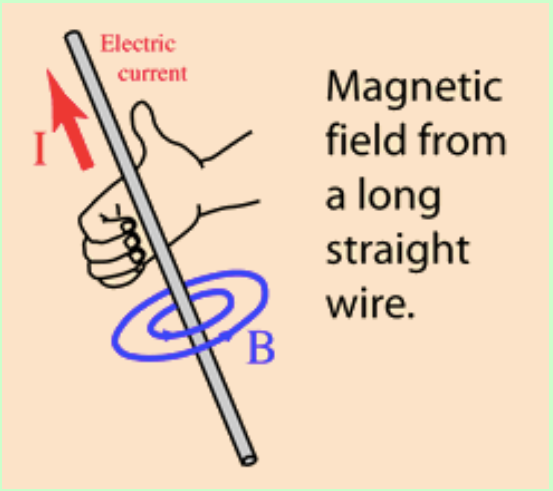
1. Faraday's Law



$$\Delta V = \vec{E}d = -N\frac{\Delta(\vec{B}A)}{\Delta t}$$

$$\boxed{\frac{\Delta\vec{B}}{\Delta t} \rightarrow \vec{E}}$$

2. Ampere's Law



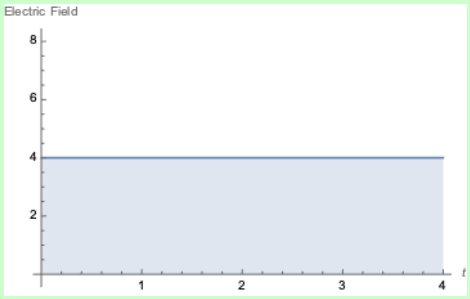
$$\vec{B}d = \mu_0 I + \mu_0\epsilon_0\frac{\Delta(\vec{E}A)}{\Delta t}$$

$$\boxed{\frac{\Delta\vec{E}}{\Delta t} \rightarrow \vec{B}}$$

For light to form, need source to be **oscillating**… (Consider a single charge that produces)

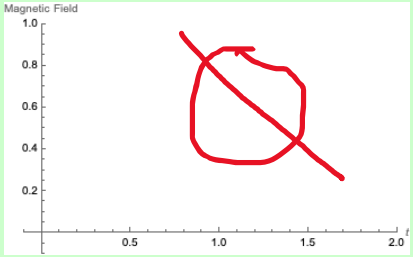
1. Stationary E Field

E



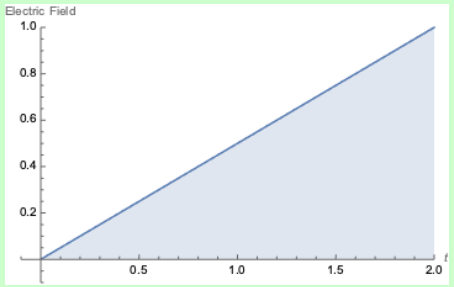
Take slope

B



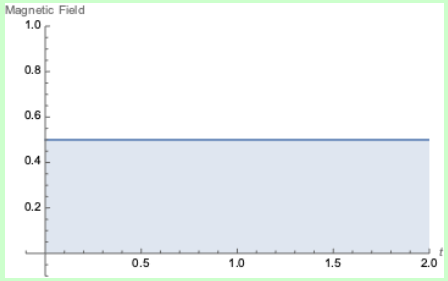
2. Linear E Field

E



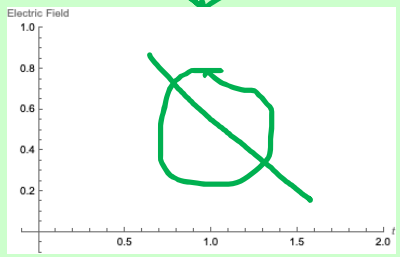
Take slope

B



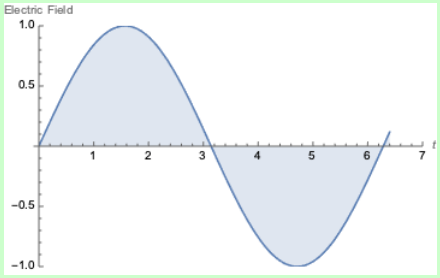
Take slope

E



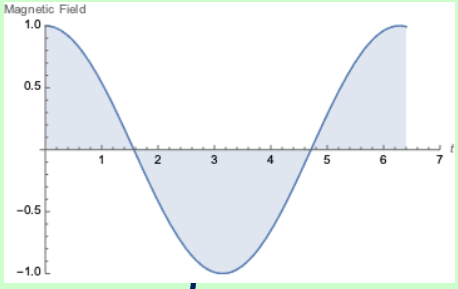
3. Sinusoidal E Field

E



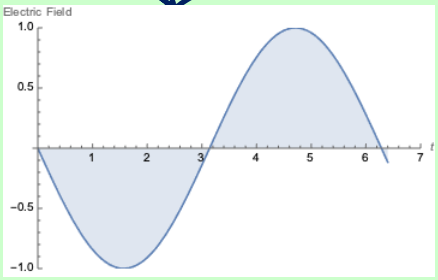
Take slope

B



Take slope

E



So, light has to be a Sin wave, with perpendicular E and B that are in phase

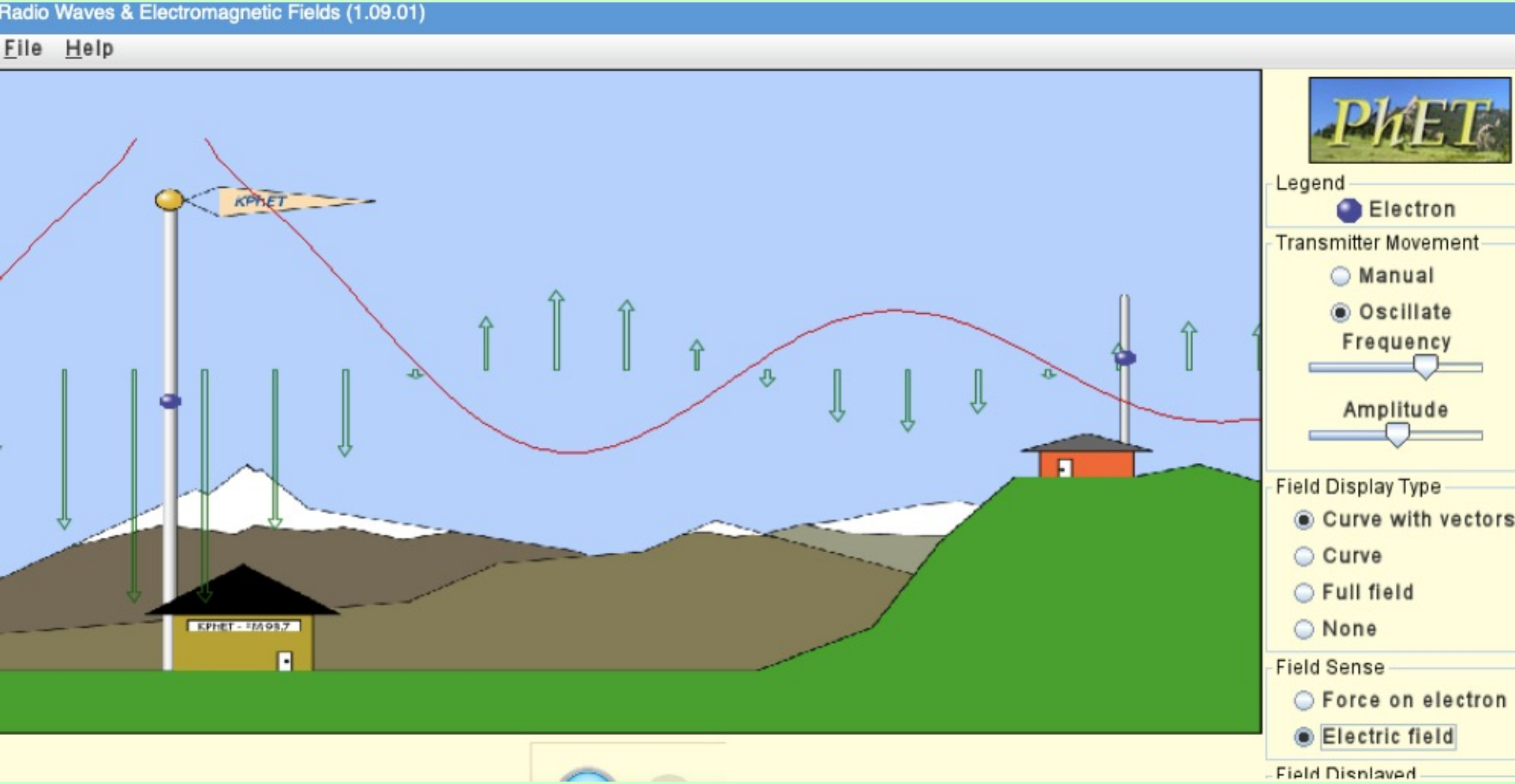
$$E(x,t) = E_0 \sin(kx - \omega t + \phi_0)$$
$$B(x,t) = B_0 \sin(kx - \omega t + \phi_0)$$

Ex. **A.** Find the wavelength and frequency, and show that v = c

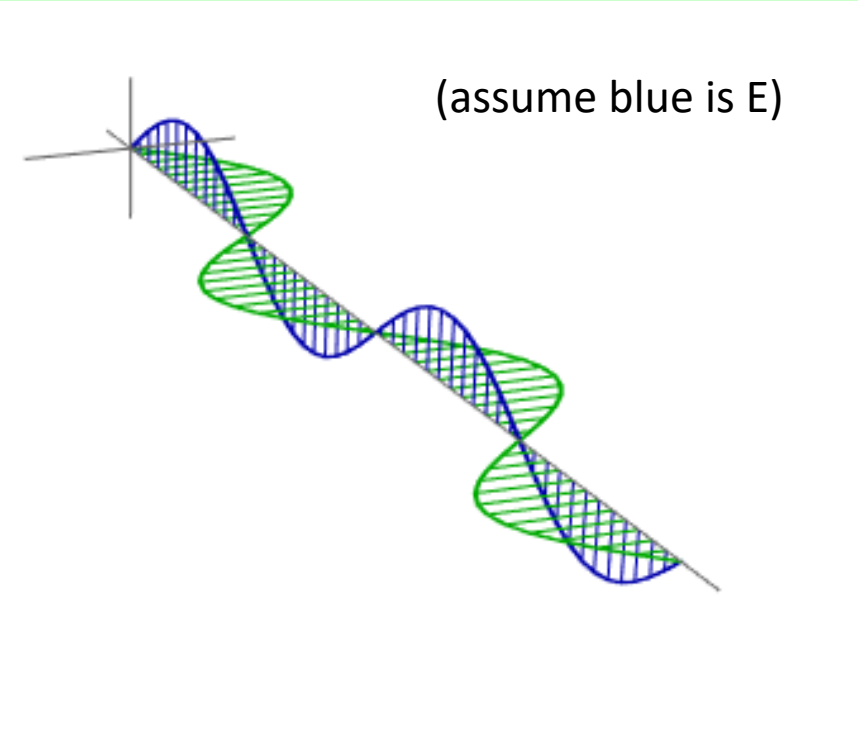
$$E(x,t) = 5.35 \sin(1.39 \times 10^7 x - 4.17 \times 10^{15} t)$$

B. What is B(x,t)?

Great demo


Radio Waves & Electromagnetic Fields (1.09.01)
File
Help
PhET
Legend
Electron
Transmitter Movement
Manual
Oscillate
Frequency
Amplitude
Field Display Type
Curve with vectors
Curve
Full field
None
Field Sense
Force on electron
Electric field
Field Displayed

Ex. Which direction is this light wave propagating?



A. Into board our out?

B. If |E| = 50 N/C, what is |B| ?

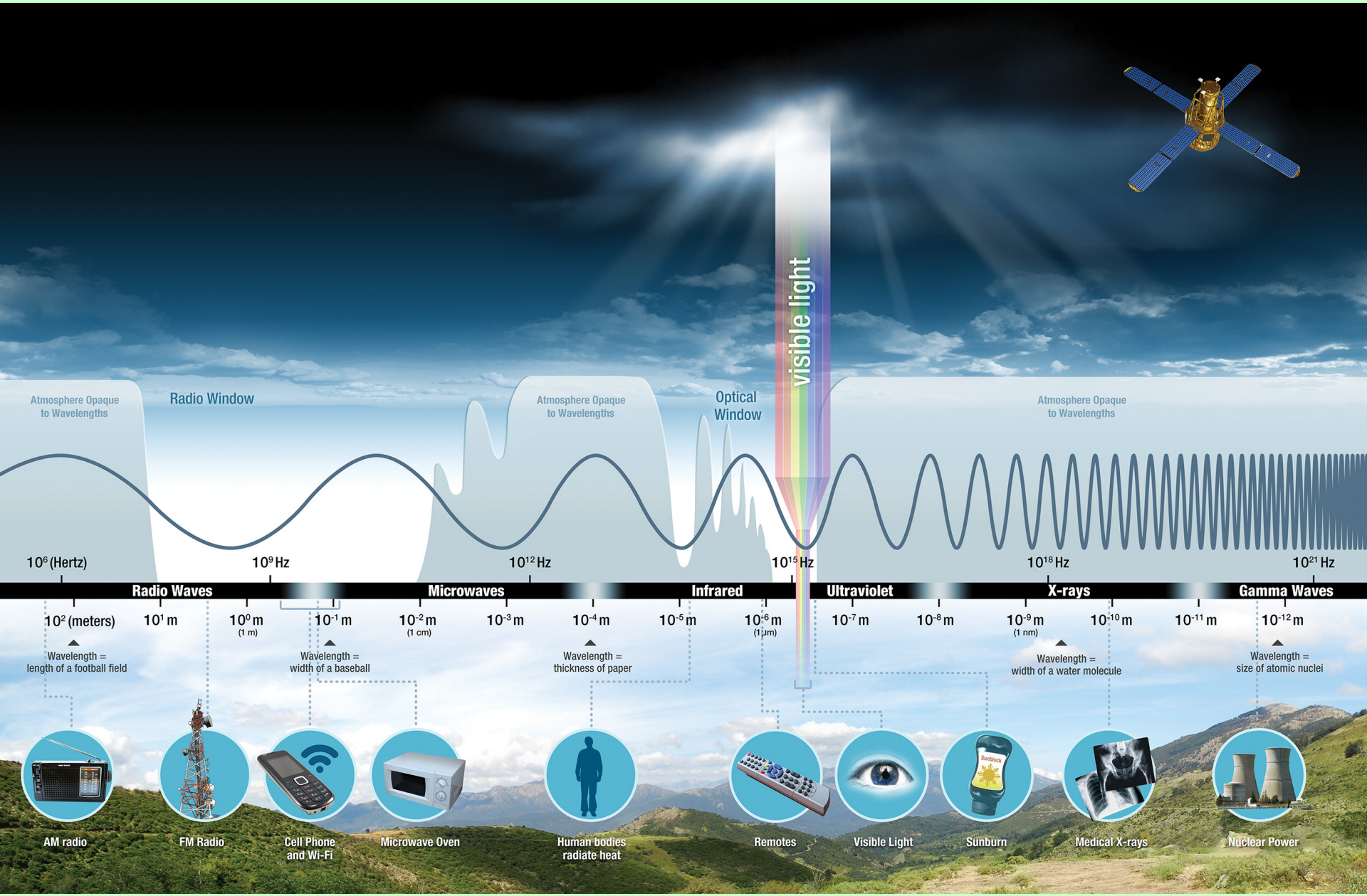
C. If $\lambda = 450\ \mathrm{nm}$ what is f?

Ex. What wavelength is a radio wave of FM 102.7 MHz?

EM Spectrum

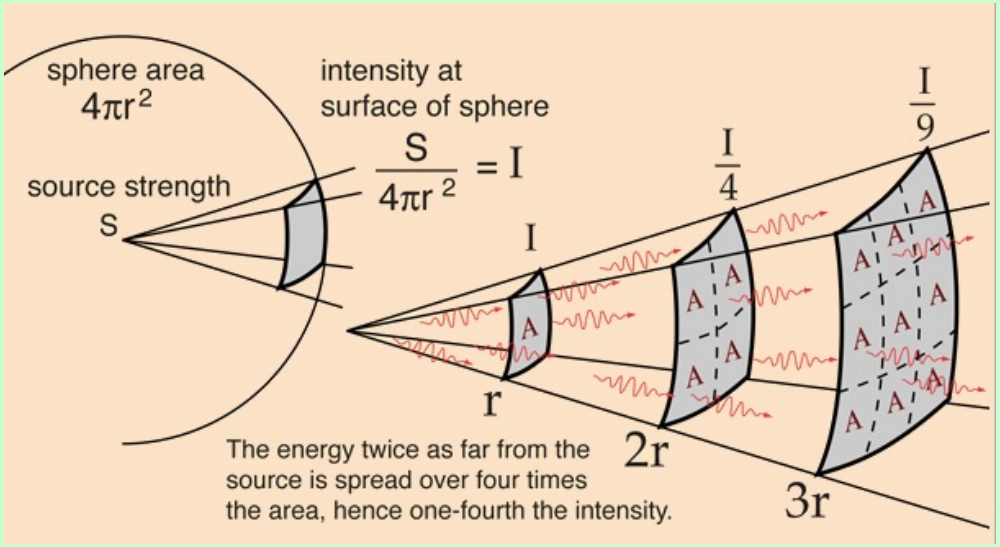
Visible: 400 – 800 nm

f?k?





Ripples: Power spreads out over $2\pi r$



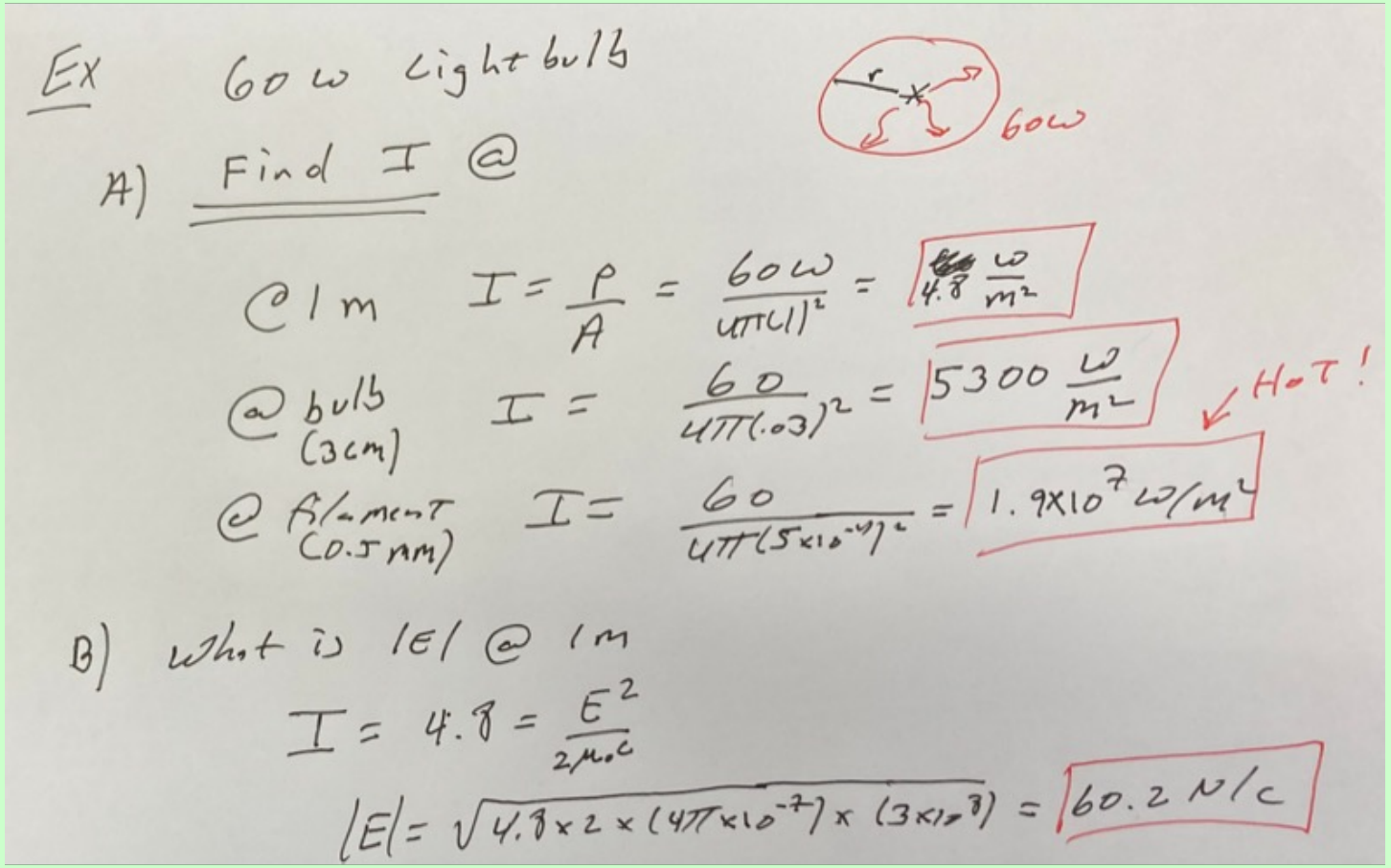
Intensity: Power spreads over surface area of sphere

$$I = \frac{\text{Power}}{\text{Area}} = \boxed{\frac{P}{4\pi r^2}}$$

$$I = \langle \vec{S} \rangle = \frac{1}{2} \frac{E^2}{c\mu_0}$$

Ex. **A.** Find the Intensity @ 1m, 3cm(bulb), 0.5mm(filament) from a 60 W bulb

B. Find the |E| at 1m



Radiation Pressure:

Demo: radiometer

Maxwell



$$\Delta \vec{p} = \frac{E_{absorbed}}{c}$$

$$\vec{F} = \frac{\Delta \vec{p}}{\Delta t} = \frac{1}{c}\frac{\Delta E}{\Delta t} = \frac{Power}{c}$$

$$P_{rad} = \frac{F}{A} = \frac{Power/c}{A} = \frac{I}{c}$$

$$\boxed{P_{rad} = \frac{I}{c}}$$ Radiation pressure

Ex. The solar Intensity at Earth is I = 1300 W/m^2. What size solar sail would accelerate at 0.01 m/s^2

c.f. Sec 1.2 pg 4